Advanced Dual-Process Cognitive Architectures:
Unifying Geometric Persistence, Semantic Elasticity, and Zero-Trust Edge Orchestration

Dane Robert Jones
ORCID: 0009-0003-8014-7298
Software Engineering Student (WGU) & Technical SME
djo4082@wgu.edu | danerjones@gmail.com

April 2, 2026

Application #64/026,345
Confirmation #6582
Patent center #75131893
Received 04/02/2026 4:28:41 PM

Abstract

The deployment of autonomous artificial intelligence systems and agentic orchestration frameworks on localized, computationally constrained edge hardware introduces a complex matrix of geometric, semantic, and physical challenges. Traditional retrieval-augmented paradigms, which rely predominantly on dense vector spaces and static proximity metrics, are mathematically ill-equipped to facilitate the complex, cross-domain analogical thinking required for autonomous problem-solving, resulting in a failure state defined as "Domain Tunneling."

Furthermore, edge deployments constrained by strict physical limits—such as memory bandwidth, thermal design power limits, and variable random access memory capacity of consumer-grade inference accelerators—are inherently vulnerable to context bloat, sequence scaling bottlenecks, and adversarial resource exhaustion attacks. Hardware analysis of contemporary entry-level inference accelerators indicates that heavily restricted memory configurations and narrowed bus widths render them profoundly inadequate for housing a massive language model alongside a continuously expanding, unstructured conversational context. A single, complex multi-turn generative conversation coupled with multi-agent tool-call results rapidly exhausts this capacity, triggering catastrophic system failures. This paper introduces a dual-process architecture designed to prevent AI "burnout" and "tunnel vision" by splitting labor between a high-speed execution layer (the Imperative Shell) and a deep-thinking semantic layer (the Functional Core).

1. The Pathology of Standard Retrieval Paradigms and Domain Tunneling

The contemporary AI paradigm treats knowledge bases as static, unstructured repositories where queries are embedded into dense, high-dimensional vector spaces. The system retrieves text chunks based exclusively on nearest-neighbor proximity, utilizing cosine similarity as the ubiquitous metric. Defined mathematically as the dot product of two vectors divided by the product of their magnitudes, cosine similarity is computationally inexpensive and highly effective for direct factual recall. However, when applied to autonomous reasoning agents, this metric inadvertently induces the phenomenon of **Domain Tunneling**.

In high-dimensional vector embeddings, semantically related terms naturally cluster together into dense conceptual neighborhoods. Conversely, concepts from entirely disparate academic or practical domains—even if they share profound structural, logical, or analogical isomorphisms—are mapped to distant coordinates. Because standard retrieval algorithms enforce strict similarity thresholds to prevent context pollution, the system simply cannot bridge these vast semantic gaps. **Why this matters:** Genuine innovation requires lateral, cross-disciplinary extrapolation. By fragmenting documents into arbitrary token lengths and discarding the topological and relational metadata connecting various concepts, critical multi-hop reasoning paths are destroyed. An autonomous agent tasked with optimizing a complex distributed network might benefit immensely from a structural analogy drawn from fluid dynamics or traffic engineering; however, a standard retrieval system will never recall the fluid dynamics documentation because its cosine similarity to the network engineering query falls below the acceptable inclusion threshold. Our architecture replaces this rigid "closeness" with a flexible geometric bridge, allowing the AI to discover structural isomorphisms across distant fields.

2. Dual-Process Cognitive Mapping and Abstraction of Hardware Layers

To solve the "How" of this cognitive split, we model operations on established "Dual-Process" cognitive theories found in human psychology: System 1 (fast, intuitive) and System 2 (slow, deliberate). By physically and algorithmically decoupling routine semantic recall from complex analogical reasoning, the architecture preserves strict cognitive economy while enabling high-fidelity, cross-domain serendipity.

2.1 The Imperative Shell: Constrained Edge Inference

The Imperative Shell resides exclusively on the system's primary graphical accelerator (GPU). It is mathematically forbidden from executing any state mutation logic, variance calculations, vector space warping, or database hygiene operations. **The Why:** By reserving computational cores exclusively for constant-time $\mathcal{O}(1)$ immediate generation of text and logic based on pre-assembled prompts, we ensure the user gets an instant response. The Imperative Shell acts as the conscious, deliberate executor, utilizing strict Top-k retrieval combined with low-temperature generative settings to guarantee absolute factual adherence while preventing generative hallucinations.

2.2 The Functional Core: Reconfigurable Logic Processing

Conversely, the heavy computational lifting regarding geometric context management is completely offloaded to a Functional Core, residing on a localized reconfigurable logic processor (e.g., an FPGA). This layer executes the "Subconscious Loop" (System 2), operating entirely asynchronously to cast a wide semantic net across thousands of disparate domains. **The How:** To facilitate this split without inducing crippling latency, the architecture relies on high-speed peripheral interconnect bifurcation algorithms and direct memory mapping utilizing Peer-to-Peer Direct Memory Access (P2P DMA) controllers. This is strictly required to prevent the host CPU from becoming a severe latency bottleneck during high-bandwidth tensor transfers between the reconfigurable logic processor and the graphical accelerator.

2.3 Consumer-Grade Emulation and Hybrid Cloud Offloading

For devices lacking dedicated FPGAs, the architecture leverages Advanced Vector Extensions (AVX-512) on the host CPU to process tensor operations directly via Single Instruction, Multiple Data (SIMD) processing. For devices strictly bound by thermal or battery constraints, a hybrid "Local-First" cloud-offloading paradigm is supported: the Imperative Shell (System 1) executes locally

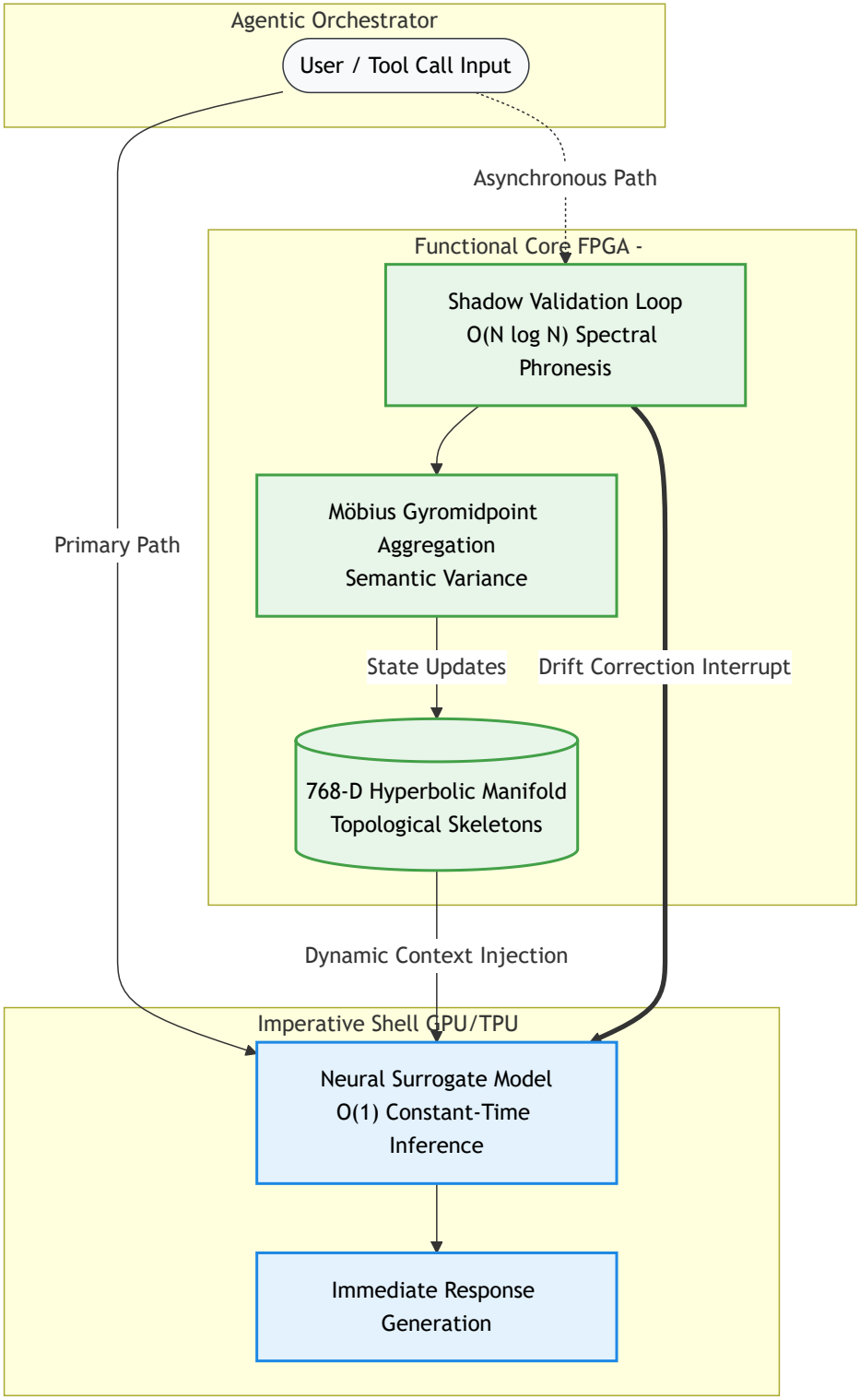

Figure 1: The Heterogeneous Hardware Labor Split, demonstrating the physical bifurcation of the $\mathcal{O}(1)$ Neural Surrogate Model and the $\mathcal{O}(N \log N)$ Shadow Validation Loop.

to preserve privacy and deterministic latency, while the heavy topological graph traversals and semantic variance modeling of the Functional Core (System 2) are securely and asynchronously offloaded to elastic cloud infrastructure.

2.4 External PCIe Expansion and Active Switching

For prosumer deployments where host mainboards fundamentally lack the BIOS-level support required for explicit PCIe lane bifurcation, the physical requirements can be satisfied via External PCIe Expansion Enclosures equipped with an **integrated active PCIe switch** (e.g., a PLX or Broadcom switch microchip). This topology permits true Peer-to-Peer Direct Memory Access (P2P DMA) to occur entirely within the expansion chassis, completely bypassing host CPU and mainboard chipset bottlenecks, ensuring absolute physical isolation of thermal and power loads.

3. Semantic Elasticity and Dynamic Attention Gating

A major challenge for local AI is "context bloat," which exhausts memory and triggers instruction decay. We solve this through **Attention Gating**, which functions as a focus dial governed by a Focus Coefficient (ϕ), derived in real-time from the semantic variance of the conversational trajectory.

The Innovation: In a high-dimensional vector space, mapping complex hierarchical data requires a Riemannian manifold where the Fréchet mean lacks a closed-form solution. Standard iterative approximation algorithms like Riemannian gradient descent induce massive $\mathcal{O}(N^2)$ latency spikes. Instead, the architecture relies entirely on the **Möbius Gyromidpoint** aggregation algorithm, providing a deterministic, closed-form expression to compute geometric averages in strict $\mathcal{O}(N)$ constant time:

$$\mathsf{AGG}(\mathbf{x}_i; \alpha) = \frac{1}{2} \otimes_\kappa \left(\frac{\sum_{i=1}^{N} \alpha_i \gamma_i \mathbf{x}_i}{\max\left(\epsilon, \sum_{j=1}^{N} |\alpha_j|(\gamma_j - 1)\right)} \right) \tag{1}$$

Once the hyperbolic semantic centroid (μ) is established, the Semantic Variance (V_s) is quantified as the average squared distance of each individual conversational turn from this central point:

$$V_s(C) = \frac{1}{N} \sum_{i=1}^{N} d^2(v_{t-i}, \mu) \tag{2}$$

During focused work, ϕ approaches zero to gate out tangential injections; conversely, when a user poses abstract queries, ϕ spikes to trigger a review mode, actively injecting structural analogies discovered by the subconscious loop.

4. Hyperbolic Geometry and Geometric Context Persistence

Standard transformer architectures rely on Euclidean vector spaces that fundamentally struggle to accurately map complex, hierarchical relational data without severe distortion. We project all contextual embeddings into a 768-dimensional **Poincaré ball** ($\mathbb{D}^{768}$).

4.1 Riemannian Repulsion and Hard Geometric Clipping

Hyperbolic geometry intrinsically possesses an exponential expansion of space relative to its radius, rendering it vastly superior for embedding tree-like structures and knowledge graphs

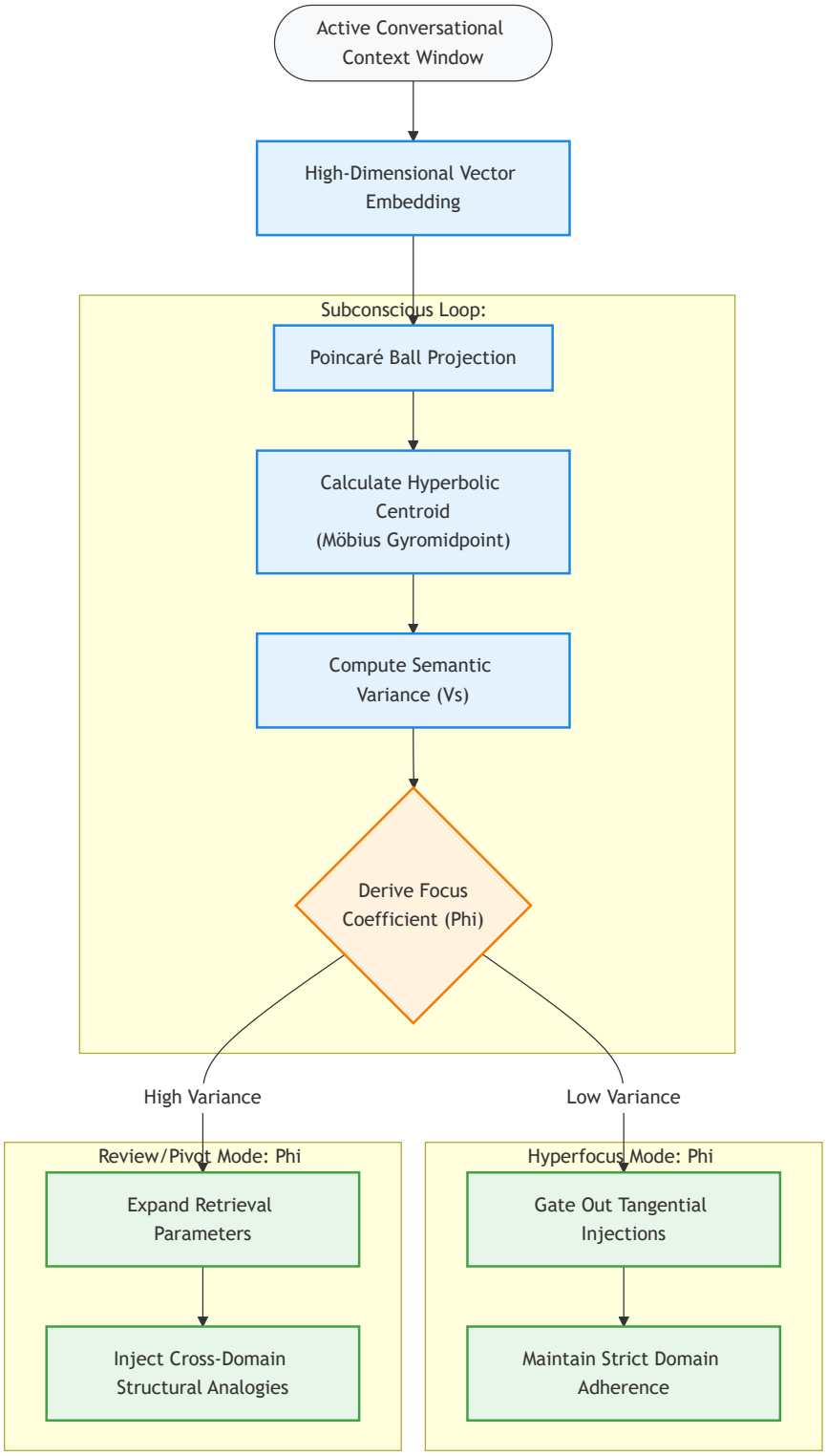

Figure 2: Dynamic Attention Gating Algorithm, detailing the derivation of the Focus Coefficient (ϕ) via Möbius Gyromidpoint Aggregation.

with low distortion. To prevent "Centroid Collapse," where distinct semantic peaks merge into an indistinguishable mass, the architecture enforces **Riemannian Repulsion**, ensuring distinct embeddings maintain identified boundaries:

$$\mathcal{R}(v_{new}) = \sum_i \frac{\nabla d_{\mathbb{D}}(v_{new}, v_i)}{\exp(d_{\mathbb{D}}(v_{new}, v_i))} \tag{3}$$

To ensure hardware stability and prevent "Sybil Repulsion" compute exhaustion attacks, the architecture implements constant-time $\mathcal{O}(1)$ **Hard Geometric Clipping** to truncate vector trajectories before they reach the absolute boundary. For a manifold with defined curvature c, the system applies:

$$clip_c(v) = \begin{cases} v & \text{if } \|v\| < \frac{1}{\sqrt{c}} \\ \left(\frac{1}{\sqrt{c}} - \epsilon\right) \frac{v}{\|v\|} & \text{otherwise} \end{cases} \tag{4}$$

Table 1: Hyperbolic Space Management Techniques

Management Technique	**Algorithmic Mechanism**	**Computational Complexity**	**Security Implication**
Beltrami-Klein Transition	Secondary manifold projection mapping	$\mathcal{O}(N)$ with high floating-point overhead	Vulnerable to Sybil Repulsion compute exhaustion attacks
Hard Geometric Clipping	Static boundary truncation via ϵ threshold	$\mathcal{O}(1)$ Constant Time bounded function	Mathematically neutralizes exhaustion and latency attacks

5. Context Hygiene and the Resolution of Infinite Memory Paradoxes

Unregulated context expansion remains the primary cause of out-of-memory states on constrained hardware. We implement bi-temporal modulation to resolve the paradox where infinite conversational memory must be mapped onto a strictly finite variable RAM footprint.

5.1 Adaptive Multi-Turn Exponential Decay

Boolean purges of information sever logical chains in Chain-of-Thought reasoning. Instead, we use **Multi-Turn Adaptive Exponential Decay**, where token weight (W_t) degrades based on session variance and a Bayesian Volatility Index (V_i):

$$W_{t+1} = W_t \cdot \exp\left(-\frac{1}{1 - V_i}\right) \tag{5}$$

The Why: This ensures that decay rates are context-aware. If variance remains high during complex reasoning, the decay rate decelerates, granting the model multiple turns to bridge the cognitive gap.

5.2 Topological Skeletons and Graph Compression

When tokens exceed their calculated TTL, they undergo conversion into highly compressed graph representations known as **Topological Skeletons**. These collapse functionally equivalent

interaction paths (such as redundant multi-agent navigation) into a single representative node. The raw text is discarded, but the compact graph shape is permanently retained within the hyperbolic manifold, ensuring long-term persistence without context bloat.

5.3 Structural Data Extraction and Semantic Key Restructuring

Instead of naive token chunking, the architecture implements **Semantic Key Restructuring**. This pipeline explicitly performs structure-aware chunking, treating document elements (headers, code blocks, tables) as immutable atomic units. **The Innovation:** We enrich each structural chunk using only a single generative invocation to simultaneously extract all metadata and propagate a "rolling key dictionary" from previous chunks. Finally, it restructures the database by merging disparate chunks that share identical semantic keys via constraint-based bin-packing.

Table 2: Context Extraction and Chunking Strategies

Chunking Strategy	**Fragmentation Mechanism**	**Context Preservation**	**Metadata Extraction Cost**
Standard Retrieval	Fixed-length arbitrary token slicing	High risk of semantic orphaning	High (requires multiple passes per chunk)
Semantic Key Restructuring	Immutable structural parsing	Maintained via rolling key dictionaries	Minimal (single-pass extraction protocol)

6. Idle-Time Latent Relation Mining: The Daydream Cycle

Attempting to perform deep graph traversals across massive vector databases in real-time incurs severe computational overhead. We solve this via an autonomous **Daydream Cycle**. **The Innovation:** During periods of hardware idle time, the logic processor utilizes Graph Neural Networks (GNNs) and translational embeddings to aggregate neighborhood information between isolated nodes from disjoint domains. When a high-degree structural isomorphism is found, the system generates a durable function that permanently warps the internal geometry to move those nodes closer. This ensures that cross-domain analogies function as pre-warped gravitational wells, ready for retrieval without computational latency during the active user session.

7. Zero-Trust Edge Security Mechanisms

Deploying autonomous agents on edge infrastructure requires robust mitigations against generative embedding inversion and adversarial prompt injections.

7.1 Tangent Differential Privacy against Embedding Inversion

Sentence embeddings are vulnerable to Generative Embedding Inversion Attacks (GEIA), where adversaries reconstruct original input sequences. We neutralize this via **Tangent Differential Privacy**, injecting a noise layer ($\mathcal{N}$) strictly within the tangent space:

$$v_{safe} = \exp_x(\log_x(v) + \mathcal{N}(0, \sigma^2)) \tag{6}$$

The How: The processor maps the vector to the tangent space, injects Gaussian noise, and projects it back. This microscopic permutation fundamentally breaks deterministic reconstruction pipelines while remaining bit-accurate for the model's inference loop.

7.2 Topological Entropy and Euclidean Quarantine Shards

Adversarial prompt injections rely on repetitive programmatic override commands that exhibit synthetic topological entropy signatures. The logic processor continuously monitors **Topological Entropy** of all incoming payloads. If a payload triggers the synthetic threshold, it is deterministically shunted out of the primary hyperbolic manifold and quarantined within an isolated "Euclidean Shard," neutralizing the attack before it reaches the inference engine.

7.3 VRAM Denial Mitigation via Asynchronous Queues

To prevent adversaries from intentionally executing VRAM Denial Floods, the architecture deploys a hardware-level **Asynchronous Inference Queue**. If hardware telemetry indicates critical memory capacity, the logic processor issues an absolute hardware lock to compress aged tokens into Topological Skeletons, buffering the flood of adversarial inputs in secondary memory until stability is achieved.

8. Orchestration Framework Integration and Spectral Consistency

Real-world utility depends on direct integration with existing agentic orchestration frameworks that provide models with explicit routing capabilities and tool execution.

8.1 The Localized Virtual Memory Hypervisor

Standard orchestrators store agent memories as plain-text Markdown files, incurring high disk I/O dependency. We implement a localized **Virtual Memory Hypervisor** that memory-maps designated file structures directly into the vector store. **The Why:** It intercepts memory retrieval tool calls at the virtual filesystem level to synthesize temporary, highly compressed virtual text strings from topological skeletons, eliminating disk-speed bottlenecks.

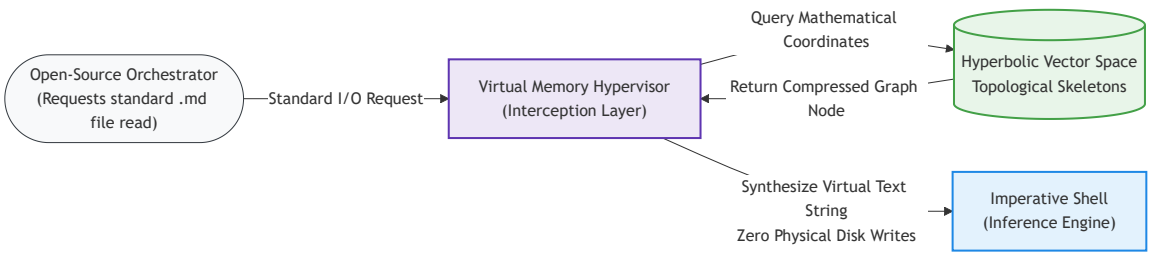

Figure 3: Virtual Memory Hypervisor Integration, illustrating the interception of disk I/O to synthesize virtual text from topological skeletons.

8.2 Cellular Sheaf Concurrency and Spectral Phronesis

When scaling orchestration layers to handle swarms of interacting agents, resolving global logical inconsistencies becomes computationally perilous. Standard methodologies rely on

exact cohomology calculations scaling at $\mathcal{O}(N^3)$. To bypass this, the architecture leverages mathematical models derived from opinion dynamics on discourse sheaves. We calculate a bit-accurate **Spectral Phronesis** index (Φ) in efficient $\mathcal{O}(N \log N)$ time:

$$\Phi = \frac{\lambda_1^+}{h_1^\epsilon + \epsilon} \tag{7}$$

This spectral heuristic utilizes the connection Laplacian and approximated dimension (h_1^ϵ) to evaluate topological obstructions. If concurrent tool calls generate contradictions, the drop in Φ allows the logic processor to isolate the conflicting cellular sheaf node and execute Selective Belief Adaptation without pausing the entire agentic swarm.

Table 3: Agentic Orchestration Integration Mechanisms

Architecture Framework	**Memory Storage Paradigm**	**Data Retrieval Mechanism**	**Synchronization Latency**
Standard Orchestrator	Physical Disk Plain-Text	File Read / Hybrid BM25 Search	High disk I/O dependency
Virtual Memory Hypervisor	Virtualized Mapped Vectors	Direct interception of protocol requests	Near-zero latency, dynamic translation

9. Addendum: Distributed Scalability and Cloud Orchestration

While built for the edge, the architecture scales via federated hierarchies. Edge nodes synchronize compressed Topological Skeletons to an overarching cloud-hosted hyperbolic manifold. This allows for a globally integrated "Synoptic Mind" that mines latent relations across millions of nodes without ever seeing raw user data. In cloud environments, these Functional Cores are distributed across **TPU Pods**, utilizing Remote Direct Memory Access (RDMA) and Matrix Multiply Units (MXUs) to parallelize the heavy tensor mathematics required for Möbius aggregations.

10. The Substrate Lag: Hardware Deficits and Adoption Vectors

Contemporary consumer chips are optimized for Euclidean math, creating a "Substrate Lag" where software cognitive architectures outpace hardware. We anticipate a transition to heterogeneous dies including native **Hyperbolic Logic Cores**, making advanced analogical reasoning a native feature of future dies. This will likely follow a phased trajectory:

- **Phase 1: Specialized Enterprise and Cloud Orchestration.** Initial adoption monopolized by enterprise-grade TPU Pods and RDMA-enabled clusters.
- **Phase 2: Prosumer Enclosures and Homelab Deployments.** Use of External PCIe Expansion Enclosures (ePCIe) acting as secondary compute nodes.
- **Phase 3: The Death of the Monolithic SoC.** Silicon fabrication shifts toward native Heterogeneous Multi-Node architectures with dedicated logic cores.

11. Conclusion

The deployment of advanced cognitive architectures on localized hardware requires an evolution beyond standard linear retrieval. By physically bifurcating deterministic action from geometric

memory management across heterogeneous processing layers, this unified framework definitively redefines the security, efficiency, and scale of artificial intelligence operations at the extreme edge.

References

Akai, I. M., & Uford, I. C. (2025). Managing academic research output through ResearchGate: Evidence from federal universities in south-south Nigeria. *Journal of Economics, Innovative Management and Entrepreneurship*, *3*(3). https://doi.org/10.59652/jeime.v3i3.604

arXiv Daily. (2026, March 26). *Machine Learning 2026_3_26*. Academic Express. http://www.arxivdaily.com/thread/78115

Bayat, S. (2026). *Spectral sheaf heuristics for consistency detection in multi-agent systems*. Research Square. https://doi.org/10.21203/rs.3.rs-8830176/v1

Buneman, P., Christie, G., Davies, J. A., Dimitrellou, R., Harding, S. D., Pawson, A. J., ... & Wu, Y. (2020). Why data citation isn't working, and what to do about it. *Database*, *2020*, baaa022. https://doi.org/10.1093/database/baaa022

Chami, I., Gu, A., Chatziafratis, V., & Ré, C. (2019). Hyperbolic deep neural networks: A survey. *arXiv preprint arXiv:2101.04562*. https://arxiv.org/pdf/2101.04562

Kling, R., & McKim, G. (1999). Scholarly communication and the continuum of electronic publishing. *Journal of the American Society for Information Science*, *50*(10), 890–906. https://doi.org/10.1002/(sici)1097-4571(1999)50:10<890::aid-asi6>3.0.co;2-8

Li, H., Xu, M., & Song, Y. (2023). Generative embedding inversion attack to recover the whole sentence. *Findings of the Association for Computational Linguistics: ACL 2023*, 13919–13931. https://aclanthology.org/2023.findings-acl.881.pdf

Li, M., Zhang, P., Xing, W., Liu, Z., Groth, P., & Worring, M. (2025). *Using large language models to tackle fundamental challenges in graph learning: A comprehensive survey*. arXiv. https://arxiv.org/html/2505.18475v1

Lou, A., Lim, I., Isola, P., Sra, S., & Solomon, J. (2020). Differentiating through the Fréchet Mean. *Proceedings of the 37th International Conference on Machine Learning*, *119*, 6513–6524. http://proceedings.mlr.press/v119/lou20a/lou20a-supp.pdf

Mangla, B. (2026). *MDKeyChunker: Single-call LLM enrichment with rolling keys and key-based restructuring for high-accuracy RAG*. arXiv. https://arxiv.org/html/2603.23533v1

Math Stack Exchange. (2023). *Are the same of Fréchet mean and Möbius gyrocentroid in the Poincaré ball model?* https://math.stackexchange.com/questions/4532796/

Poudel, B. (2026). *How OpenClaw Works: Understanding AI Agents Through a Real Architecture*. Medium. https://bibek-poudel.medium.com/how-openclaw-works-understanding-ai-agents-thro

Ranga, S., Mao, R., Cambria, E., & Chattopadhyay, A. (2026). *The plagiarism singularity conjecture*. ResearchGate. https://www.researchgate.net/publication/392503617

Sgantzos, K., Stelios, I., Tzavaras, N., & Theologou, M. (2024). Minds and machines: Evaluating the feasibility of constructing an advanced artificial intelligence. *Discover Artificial Intelligence*, *4*(1), 58. https://doi.org/10.1007/s44163-024-00216-2

Shanks, J., & Arlitsch, K. (2016). Making sense of researcher services. *Journal of Library Administration*, *56*(3), 295–316. https://doi.org/10.1080/01930826.2016.1146534

Stall, S., Bilder, G., Cannon, M., Chue Hong, N., Edmunds, S., Erdmann, C. C., … & Townsend, R. (2023). Journal production guidance for software and data citations. *Scientific Data*, *10*(1), 656. https://doi.org/10.1038/s41597-023-02491-7

Tsai, Y.-C., Hsiao, H., Chen, K.-Y., & Lin, S.-D. (2026). *Concept-aware privacy mechanisms for defending embedding inversion attacks*. arXiv. https://arxiv.org/html/2602.07090v1

VoltAgent. (2026). *awesome-openclaw-skills* [GitHub repository]. https://github.com/VoltAgent/awesome-openclaw-skills

Wang, B. (n.d.). *Bei Wang - Scientific Computing and Imaging Institute*. University of Utah. https://www.sci.utah.edu/~beiwang/

Wang, Y., Liu, Y., & Derr, T. (2024). *Edge classification on graphs: New directions in topological imbalance*. ResearchGate. https://www.researchgate.net/publication/381510788

Wang, Y., et al. (2024). GGBall: Graph generative model on Poincaré ball. *OpenReview*. https://openreview.net/forum?id=4zRRnDscqn

Wren, J. D., & Georgescu, C. (2022). Detecting anomalous referencing patterns in PubMed papers suggestive of author-centric reference list manipulation. *Scientometrics*, *127*(10), 5753–5771. https://doi.org/10.1007/s11192-022-04503-6

Ying, L. (2024). Tangent differential privacy. *Journal of Machine Learning*. https://arxiv.org/html/2406.04535v1

Zhang, X., Han, Y., Qi, J., Xie, M., Wu, Z., & Dai, X. (2026). *WebNavigator: Global web navigation via interaction graph retrieval*. arXiv. https://arxiv.org/html/2603.20366v1